

1 · WHY BIRDS ARE DIFFERENT ★★★

THE SINGLE MOST IMPORTANT AVIAN FACT The bird's granulocyte is the **HETEROPHIL**, not the neutrophil. It has **rod- or spindle-shaped eosinophilic granules** and, critically, it **LACKS MYELOPEROXIDASE and lysozyme**. Therefore avian purulent inflammation **cannot liquefy the debris** and forms a **DRY, FIRM, CASEOUS (inspissated) mass instead of flowing pus**. This is why the "abscess" of a bird is always a **caseous nodule or plug** — in the trachea, air sac, or joint.

Feature	Bird
Red cells	Nucleated and oval ; THROMBOCYTES replace platelets (also nucleated)
Lymph nodes	ABSENT in chickens . Instead: BURSA OF FABRICIUS (B cells) , THYMUS (T cells) , caecal tonsils, Peyer's patches, Harderian gland, diffuse lymphoid aggregates
Nitrogen waste	URIC ACID (uricotelic) — poorly soluble, so any renal failure or dehydration precipitates it as GOUT
Kidney	Has a RENAL PORTAL SYSTEM — drugs injected in the leg reach the kidney first
Air sacs	Thin and almost avascular — so inflammation there is poorly cleared and airsacculitis persists as caseous sheets
Other	No diaphragm , no sweat glands (heat stress matters), medullary bone as a calcium store in layers

2 · RANIKHET DISEASE (NEWCASTLE DISEASE) ★★★

- **Avian paramyxovirus-1 (APMV-1)**, Paramyxoviridae. **Notifiable**. The most important poultry disease of India
- **Pathotypes by virulence: LENTOGENIC** (mild — **F1, B1, LaSota** vaccine strains) · **MESOGENIC** (moderate — **R2B "Mukteswar"**, Komarov, used as boosters) · **VELOGENIC** (severe — **viscerotropic VVND and neurotropic**)

THE THREE LESIONS THAT NAME IT ★ ① **HAEMORRHAGE / NECROTIC ULCERS ON THE TIPS OF THE PROVENTRICULAR GLANDS** — pinpoint red dots in a row at the base of the papillae ② **NECROSIS AND ULCERATION OF THE CAECAL TONSILS AND PEYER'S PATCHES** — raised, "boat-shaped" or button-like ulcers along the intestine ③ **HAEMORRHAGIC TRACHEITIS** and congestion of the whole respiratory tract

Also: greenish watery diarrhoea, gasping, **NERVOUS SIGNS** — **torticollis (twisted neck)**, **paralysis of legs and wings**, **opisthotonus**, sudden drop in egg production with **soft-shelled and depigmented eggs**, ovarian follicle haemorrhage and rupture (**yolk peritonitis**).

Micro: non-suppurative encephalomyelitis — **perivascular cuffing**, **neuronal degeneration**, **gliosis**. **Diagnosis:** HA/HI test, virus isolation in embryonated eggs, RT-PCR.

3 · IBD (GUMBORO) & CHICKEN INFECTIOUS ANAEMIA ★★

Disease	Agent & lesions
INFECTIOUS BURSAL DISEASE (Gumboro)	Birnavirus (double-stranded RNA, two segments); extremely resistant in the environment . Affects chicks of about 3–6 weeks , when the bursa is at its largest. Target = BURSA OF FABRICIUS: first swollen to 2–3× normal , oedematous , with a yellow gelatinous transudate on the serosa , often haemorrhagic ("strawberry bursa") ; then by about day 8 it ATROPHIES and becomes small and grey. "PAINT-BRUSH" HAEMORRHAGES in the thigh and pectoral muscles are the classic second lesion; also nephrosis with urate, and a dehydrated darkened carcass. The real damage is IMMUNOSUPPRESSION — B-cell destruction means vaccination failures against Ranikhet and other diseases
CHICKEN INFECTIOUS ANAEMIA	Gyrovirus , family Circoviridae; vertically transmitted; young chicks about 2–4 weeks. Destroys haemocytoblasts in the bone marrow and T-cell precursors in the thymus . APLASTIC ANAEMIA — pale, fatty, yellow-white bone marrow (normally red), PCV falls below about 27% , pale carcass, THYMIC ATROPHY , haemorrhages in muscle and subcutis. With secondary <i>Clostridium</i> it produces "BLUE WING DISEASE" / gangrenous dermatitis of the wing tips. Strongly immunosuppressive

4 · IB AND ILT — THE TWO RESPIRATORY VIRUSES ★★

	INFECTIOUS BRONCHITIS	INFECTIOUS LARYNGOTRACHEITIS
Agent	Coronavirus (many serotypes, poor cross-protection)	Gallid herpesvirus-1
Severity of the cough	Gasping, sneezing, tracheal rales, watery eyes; milder	SEVERE dyspnoea — the bird extends its neck and gasps, and COUGHS UP BLOOD-STAINED MUCUS
Trachea	Serous then caseous plug in trachea/bronchi	Haemorrhagic tracheitis with BLOOD-STAINED CASEOUS CASTS that can suffocate the bird
Other systems	Nephropathogenic strains → interstitial nephritis, swollen pale kidneys, urate deposits and gout . Reproductive: chicks infected in the first 2 weeks suffer permanent oviduct damage → "FALSE LAYER" ; adults lay misshapen, soft, wrinkled, rough-shelled eggs with WATERY ALBUMEN	Conjunctivitis, sinusitis; egg drop
Micro	Loss of cilia, epithelial deciliation and hyperplasia	SYNCYTIAL giant cells with COWDRY TYPE A INTRANUCLEAR inclusions in the tracheal epithelium — diagnostic

5 · FOWL POX & AVIAN INFLUENZA

Disease	Key facts
FOWL POX	Avipoxvirus ; spread by MOSQUITOES and through skin abrasions — hence seasonal. CUTANEOUS (dry) form — nodular wart-like scabs on the comb, wattles, eyelids and other unfeathered skin ; usually mild, self-limiting. DIPHTHERITIC (wet) form — yellow-white CASEOUS plaques firmly adherent to the mouth, pharynx, larynx and trachea , which peel off leaving a bleeding ulcer; this form kills by suffocation and carries much higher mortality. Micro : epidermal hyperplasia with ballooning degeneration and large INTRACYTOPLASMIC BOLLINGER BODIES containing the elementary Borrel bodies
AVIAN INFLUENZA	Influenza A ; HPAI strains such as H5N1 and H5N8 — notifiable and zoonotic . Wild waterfowl are the reservoir. Sudden very high mortality, often before signs are noticed . Cyanosis and oedema of the comb and wattles , subcutaneous oedema of the head and neck, haemorrhages on the shanks and feet ("red hocks") , petechiae on the proventriculus, epicardium and abdominal fat , and necrotic foci in the pancreas and spleen . Severe egg drop. Low-pathogenic strains cause only mild respiratory disease and egg drop

6 · MAREK'S DISEASE VS LYMPHOID LEUKOSIS ★★★

This comparison is asked almost every year. Learn the table, not the paragraphs.

	MAREK'S DISEASE	LYMPHOID LEUKOSIS
Virus	Gallid herpesvirus-2 (a DNA herpesvirus)	Avian leukosis virus (an RNA retrovirus)
Cell type	T lymphocyte	B lymphocyte
Age	Younger — usually 6 weeks to 6 months , can be earlier	Older — rarely before 16 weeks
PERIPHERAL NERVES	ENLARGED — sciatic and brachial plexus become 2–3× thick, greyish-yellow, oedematous, with LOSS OF THE NORMAL CROSS-STRIATIONS ; the enlargement is ASYMMETRICAL . This is the single best gross criterion	NEVER involved
BURSA of Fabricius	Usually atrophied ; diffuse infiltration if involved	ALWAYS involved — NODULAR tumour in the bursa ; the bursa is the site of origin
Eye	"GREY EYE" / ocular form — depigmented iris and an irregular, pinpoint, non-responsive pupil	Not affected
Skin	Cutaneous form — enlarged feather follicles , the source of infectious dander	Not affected
Classic sign	"RANGE PARALYSIS" — the bird sits with one leg stretched forward and the other backward	Non-specific wasting, pale comb, distended abdomen
Spread	HORIZONTAL , by inhaling DANDER from the feather follicle epithelium — the only site where fully infectious virus is produced. Highly contagious, dust survives for months	Mainly VERTICAL (through the egg) and by close contact; slow spread
Control	VACCINATION WORKS — HVT (turkey herpesvirus) given in ovo or at day old	No vaccine — control by eradication of infected breeders

ONE-LINE MEMORY HOOK MAREK = M for Motor nerves and Marek's is the younger bird. LEUKOSIS = L for Later (16 weeks) and Liver/bursa Lumps, with nerves Left alone.

7 · RETICULOENDOTHELIOSIS · AVIAN ENCEPHALOMYELITIS · AVIAN NEPHRITIS

Disease	Key point
Reticuloendotheliosis	Retrovirus (REV) , unrelated to ALV. Produces a runting–stunting syndrome with abnormal feathering, plus lymphoma of liver, spleen and other viscera . Because it can also enlarge peripheral nerves , it is the one condition that genuinely mimics Marek's disease. Also a common contaminant of vaccines
AVIAN ENCEPHALOMYELITIS ("epidemic tremor")	Picornavirus , mainly VERTICALLY transmitted ; affects chicks under about 3 weeks . Ataxia, sitting on hocks, progressive paresis, and fine rapid TREMORS of the HEAD and NECK — obvious when the bird is held. Adults show only a transient egg drop; survivors may develop cataract . There are no useful gross lesions . Micro : non-suppurative encephalomyelitis with CENTRAL CHROMATOLYSIS of neurons , perivascular cuffing, and LYMPHOID NODULES IN THE MUSCULAR LAYER OF THE GIZZARD and proventriculus — this is the diagnostic histological lesion
Avian nephritis	Astrovirus ; young chicks. Interstitial nephritis with pale swollen kidneys, urate deposition and visceral gout , growth retardation and runting. Often subclinical but adds to other nephropathies

8 • IBH AND HYDROPERICARDIUM SYNDROME ★★ (MAJOR IN INDIA)

	INCLUSION BODY HEPATITIS	HYDROPERICARDIUM SYNDROME
Agent	Fowl adenovirus (group I) , several serotypes	Fowl adenovirus serotype 4 (FAdV-4)
Other names	—	"ANGARA DISEASE" (first described at Angara Goth, Karachi, 1987) and "LITCHI HEART"
Age	Broilers about 3–7 weeks	Broilers about 3–6 weeks
Lesion	Liver enlarged, pale, "yellow-ochre", friable, with focal necrosis and haemorrhage; pale bone marrow and anaemia; haemorrhage in muscles	The pericardial sac is distended with clear straw-coloured JELLY-LIKE fluid (often 10–20 ml) and the heart is flabby and pale ; plus a swollen friable mottled liver with necrosis and swollen kidneys with urate
Micro	Large BASOPHILIC (sometimes eosinophilic) INTRANUCLEAR INCLUSION BODIES in hepatocytes — the common feature of both	
Mortality	Usually 10–30%, peaks in a few days	Can be very high, commonly 20–60% in an outbreak

PREDISPOSITION Both are made much worse by prior immunosuppression from IBD or chicken infectious anaemia or from aflatoxin in the feed — always look for the underlying cause.

9 • EGG DROP SYNDROME & REOVIRUS ★

Disease	Key facts
EGG DROP SYNDROME (EDS-76)	Caused by a DUCK ADENOVIRUS (Atadenovirus) — ducks and geese are the natural host and reservoir. Vertically transmitted , the virus stays latent until point of lay. The classic picture: a sudden drop in egg production of 10–40% in apparently HEALTHY birds with NO other signs. The eggs are SHELL-LESS, thin-shelled, soft, rough or DEPIGMENTED (loss of brown colour) , with watery albumen. Lesion: oedema and inflammation of the SHELL GLAND (pouch) of the oviduct , with intranuclear inclusion bodies in its epithelium; inactive ovaries and atrophic oviduct. Diagnosis by HI test
REOVIRUS infections	① VIRAL ARTHRITIS / TENOSYNOVITIS — the important one. Swelling above the hock , inflammation of the gastrocnemius and digital flexor tendon sheaths with oedema and later fibrosis; in chronic cases the GASTROCNEMIUS TENDON RUPTURES , giving a green-black bruise above the hock and permanent lameness. ② Malabsorption / runting-stunting ("helicopter disease") — poor growth, abnormal feathering, pale, undigested feed in the droppings, brittle bones. Also enteritis and hepatitis

10 • AVIAN COLIBACILLOSIS ★★★

THE TRIAD — THE SINGLE MOST-ASKED AVIAN GROSS PICTURE

Colisepticaemia produces the classic trio of **fibrinous PERIHEPATITIS** (a dull white fibrin film peeling off the liver capsule), **fibrinous PERICARDITIS** (thick opaque pericardium adherent to the heart) and **fibrinous AIRSACCULITIS** (air sacs thickened, cloudy, with yellow caseous sheets). Any one of these three in a chicken should make you say ***E. coli***.

Form	Description
Omphalitis / yolk sac infection	"MUSHY CHICK" — wet, swollen, foul-smelling navel in the first week; unabsorbed, discoloured, foul yolk sac; from dirty hatchery or floor eggs
Colisepticaemia	The triad above, plus splenomegaly and congestion; usually follows respiratory damage by Ranikhet virus, IB or mycoplasma — <i>E. coli</i> is a secondary invader
Coligranuloma (Hjärre's disease)	Firm yellow granulomas in the liver, caeca and mesentery — resembles tuberculosis but is not acid-fast
Salpingitis / egg peritonitis	Oviduct distended with laminated caseous casts; a common cause of death in layers
Swollen head syndrome, coliform cellulitis	Subcutaneous oedema of head and periorbital tissue; yellow caseous plaques under the abdominal skin

11 • INFECTIOUS CORYZA • CLOSTRIDIAL DISEASES

- **INFECTIOUS CORYZA** — *Avibacterium (Haemophilus) paragallinarum*. Marked swelling of the face and infraorbital sinus, closed swollen eyes, and a **thick, sticky nasal discharge with a characteristic FOUL, OFFENSIVE ODOUR** — the odour is the field diagnosis. Conjunctivitis, wattle oedema, sharp drop in egg production. Morbidity is high but mortality low; carriers persist in the flock
- **NECROTIC ENTERITIS** — *Clostridium perfringens* type A (and C), following mucosal damage by **coccidiosis** or a sudden feed change. The **small intestine is ballooned, thin-walled and full of dark foul gas and fluid**; the mucosa is covered by a **loose, brown-green, friable diphtheritic pseudomembrane described as a "TURKISH TOWEL"**. Sudden deaths in the best-grown birds
- **ULCERATIVE ENTERITIS ("quail disease")** — *Clostridium colinum*; deep punched-out ulcers of intestine and caeca, with pale necrotic foci in the liver
- **GANGRENOUS DERMATITIS** — *Cl. septicum* and *Cl. perfringens* entering damaged skin in birds already immunosuppressed by **chicken infectious anaemia or IBD**; dark, wet, crepitant, gangrenous skin of the wing tip and thigh
- **BOTULISM ("limberneck")** — *Cl. botulinum* type C toxin, eaten in maggots or rotting carcasses; **flaccid paralysis of the neck, wings and legs**; no gross lesion

12 • SALMONELLA INFECTIONS ★★★

	PULLORUM DISEASE	FOWL TYPHOID	PARATYPHOID
Organism	<i>Salmonella</i> Pullorum	<i>Salmonella</i> Gallinarum	<i>S. Typhimurium</i> , <i>S. Enteritidis</i> (motile)
Age	CHICKS — first 2–3 weeks	GROWERS and ADULTS	Any age, often young
Signs	Chalky WHITE DIARRHOEA that pastes and seals the VENT , huddling, gasping, high mortality	Sudden drop in production, yellow-green diarrhoea, pale shrunken comb, deaths	Enteritis, septicaemia in young birds
Lesions	Unabsorbed retained yolk sac; grey-white NECROTIC NODULES in the HEART MUSCLE, liver, lung and GIZZARD (nodules in the heart are highly suggestive); in surviving hens the ovarian ova become misshapen, discoloured, pedunculated and cystic , and these hens become the egg-transmitting carriers	Liver enlarged with a striking BRONZE / GREENISH-BRONZE discolouration , splenomegaly, miliary necrotic foci, regressed haemorrhagic ovary, catarrhal enteritis	Focal hepatic necrosis, enteritis, pericarditis; the major EGG-BORNE ZOOONOSIS of public health importance
Spread	Mainly VERTICAL through the egg — hence eradication is by testing breeders with the rapid whole-blood plate agglutination test and culling reactors		Horizontal and vertical

13 • FOWL CHOLERA • AVIAN TB • SPIROCHAETOSIS

Disease	Key facts
FOWL CHOLERA	<i>Pasteurella multocida</i> ; bipolar organisms on a Leishman/Giemsa smear of heart blood or liver. ACUTE (septicaemic): sudden death of the best birds , often found dead on the floor; cyanotic, dark purple comb and wattles; petechial and ecchymotic haemorrhages on the epicardium, serosa, abdominal fat and intestinal mucosa ; the LIVER is swollen with countless PINPOINT GREY-WHITE NECROTIC FOCI ; pneumonia is prominent in turkeys. CHRONIC : localised infection — swollen oedematous WATTLES , arthritis and swollen joints, torticollis from middle-ear and cranial infection , conjunctivitis
AVIAN TUBERCULOSIS	<i>Mycobacterium avium</i> ; a disease of older birds, backyard flocks and zoo birds — rare in short-cycle broilers. Progressive emaciation with a prominent keel despite a good appetite , pale comb, lameness. Irregular grey-white to yellow NODULES of varying size in the LIVER, SPLEEN and INTESTINE , and often in the bone marrow ; unlike mammalian TB there is little or no calcification . Ziehl-Neelsen: abundant ACID-FAST bacilli . Zoonotic risk to immunocompromised people
AVIAN SPIROCHAETOSIS	<i>Borrelia anserina</i> , transmitted by the FOWL TICK Argas persicus — so it follows tick infestation of housing. Fever, marked depression, greenish diarrhoea , progressive anaemia and paralysis . Post-mortem: greatly enlarged, mottled spleen with white foci , swollen mottled liver, pericardial effusion. Diagnosis: motile spirochaetes in a wet or Giemsa-stained blood film during the febrile phase

14 • MYCOPLASMOSIS & CHLAMYDIOSIS ★★★

Disease	Key facts
<i>Mycoplasma gallisepticum</i> = CHRONIC RESPIRATORY DISEASE (CRD)	Slow-spreading, egg-transmitted , and stress-triggered. Coughing, sneezing, tracheal rales, nasal and ocular discharge, swollen infraorbital sinus ("INFECTIOUS SINUSITIS" in turkeys) , poor growth and egg drop. Lesions : catarrhal exudate in nasal passages, sinuses and trachea; thickened, cloudy air sacs with yellow caseous beads — AIRSACCULITIS . Micro : marked LYMPHOID FOLLICLE hyperplasia in the mucosa. "COMPLICATED CRD" — when <i>E. coli</i> joins in, the perihepatitis–pericarditis–airsacculitis triad appears and mortality rises sharply. This is the commonest respiratory problem of Indian commercial layers
<i>Mycoplasma synoviae</i>	INFECTIOUS SYNOVITIS — swollen hock joints and foot pads , lameness, breast blister, pale comb; joints and tendon sheaths contain a viscous grey-yellow to creamy exudate that later becomes caseous. Also causes a subclinical airsacculitis and eggshell apex abnormalities
CHLAMYDIOSIS (psittacosis in parrots, ornithosis in other birds)	<i>Chlamydia psittaci</i> , an obligate intracellular organism; notifiable ZOOONOSIS that causes serious atypical pneumonia in people handling birds. Lesions : fibrinous AIRSACCULITIS, PERICARDITIS and PERIHEPATITIS (looks like colibacillosis), with marked enlargement of the SPLEEN and LIVER and focal necrosis; conjunctivitis and greenish diarrhoea in the live bird. Diagnosis : elementary bodies in impression smears with Giemsa, Macchiavello or Gimenez stain , plus antigen ELISA and PCR

15 • AVIAN MYCOSES

Disease	Key facts
ASPERGILLOSIS ("brooder pneumonia")	<i>Aspergillus fumigatus</i> , from mouldy litter, damp feed and contaminated hatchery/setter — the spores are inhaled in enormous numbers. Acute form in chicks — high mortality in the first two weeks, gasping without a rattle. Lesions: multiple hard, yellow-white CASEOUS NODULES (1–5 mm) scattered through the LUNGS and studding the AIR SACS ; on an exposed air sac the fungus may grow visible green-grey fuzzy sporulating colonies , which is diagnostic on sight. Micro: granulomas with septate, dichotomously branching hyphae at about 45° , giant cells; stain with PAS or Grocott's methenamine silver . Ocular and brain forms also occur. Not contagious bird-to-bird — it is an environmental disease , so the answer is always to correct the litter and feed storage
THRUSH / crop mycosis ("sour crop")	<i>Candida albicans</i> ; follows prolonged antibiotic use, poor hygiene and vitamin A deficiency . The crop mucosa is thickened and covered with a white-grey, raised, adherent pseudomembrane whose corrugated pattern is described as a "TURKISH TOWEL" ; also ulcers in the proventriculus. Birds are unthrifty with a doughy crop
FAVUS (avian ringworm, "white comb")	<i>Microsporum gallinae</i> ; a superficial dermatophytosis of the unfeathered skin of the comb and wattles , producing white, powdery, scaly, crusty patches that may thicken into wrinkled masses; feather loss on the head and neck follows. Chronic, disfiguring, rarely fatal

16 • AVIAN MYCOTOXICOSES ★★

Toxin	Target & lesion in the bird
AFLATOXIN (<i>Aspergillus flavus</i>)	Target = LIVER. DUCKLINGS are the most susceptible of all species , followed by turkey poults; chickens are relatively more resistant. Liver is enlarged, pale to yellow, firm and fatty, with marked BILE DUCT (biliary) HYPERPLASIA in chronic cases; reduced growth and feed efficiency, pale carcass with subcutaneous and muscular haemorrhage (because clotting factors fall), poor pigmentation of shanks, drop in egg production and hatchability. Profoundly IMMUNOSUPPRESSIVE — a common hidden reason for vaccine failure and for severe IBD, IBH or coccidiosis outbreaks in India. Source: damp groundnut cake and maize
OCHRATOXIN A	Target = KIDNEY — and it is more nephrotoxic to poultry than aflatoxin . Swollen, pale, mottled kidneys with distended ureters full of white urate , and VISCERAL GOUT ; polydipsia, wet litter, poor growth, poor shell quality. Also immunosuppressive
TRICHOHECENES (T-2 toxin, DON — <i>Fusarium</i>)	Direct contact necrosis wherever the toxin touches . The classic lesion is yellow-white caseous ORAL LESIONS at the COMMISSURE OF THE BEAK and on the HARD PALATE and TONGUE . Also feed refusal, gizzard erosion , intestinal necrosis, and — because trichothecenes are radiomimetic — necrosis of bone marrow and lymphoid tissue with haemorrhages , abnormal feathering and neural signs

17 • HELMINTHS OF POULTRY

Parasite	Site & pathology
Ascaridia galli	Largest nematode of poultry, in the small intestine ; heavy burdens cause enteritis, intestinal obstruction and even rupture , unthriftiness, and worms may migrate up the oviduct into the egg
Heterakis gallinarum	In the CAECA ; itself only mildly pathogenic, but it is the vector that carries Histomonas meleagridis inside its egg — this is its exam importance
Capillaria	Hair-like worms in crop, oesophagus and intestine; thickened mucosa, catarrhal or haemorrhagic enteritis, marked weight loss
Syngamus trachea ("gapeworm")	Bright red, permanently paired worms in a Y shape in the TRACHEA ; the bird "gapes" , stretches its neck and shakes its head; obstruction can suffocate young birds
Cestodes (<i>Raillietina</i> , <i>Davainea</i>)	Attach to the small intestinal mucosa; nodular enteritis, poor growth, drop in production
Trematodes (<i>Prosthogonimus</i>)	The oviduct fluke — salpingitis, thin- and soft-shelled eggs, egg peritonitis

18 · COCCIDIOSIS & HISTOMONIASIS ★★★

<i>Eimeria</i> species	Site & lesion — the species is identified by WHERE it sits
<i>E. tenella</i>	CAECA. The most acute and most fatal, especially at 3–6 weeks . Frank BLOODY DROPPINGS ; caeca are distended, thickened and full of clotted blood, which later organises into hard laminated "CAECAL CORES" . High mortality
<i>E. necatrix</i>	MID small intestine. The most damaging of the intestinal species in growers; the gut is ballooned and thickened with white spots and petechiae seen from the serosa giving a "SALT AND PEPPER" appearance
<i>E. acervulina</i>	DUODENUM. The commonest species. Transverse white bands or "LADDER-LIKE" white plaques across the duodenal mucosa, best seen from the serosal side. Poor growth rather than death
<i>E. maxima</i>	MID intestine. Thickened flaccid gut with orange-pink mucus ; the largest oocysts ; important because it damages absorption and predisposes to necrotic enteritis
<i>E. brunetti</i>	LOWER intestine, rectum and cloaca. Necrotic and haemorrhagic enteritis with a coagulated cast

HISTOMONIASIS — "BLACKHEAD" ★★★ *Histomonas meleagridis*, a flagellate protozoan. **TURKEYS** are highly susceptible (mortality can approach 100%); chickens are largely resistant carriers. It is carried inside the egg of the caecal worm *Heterakis gallinarum*, and earthworms act as transport hosts — which is why the disease persists in the soil for years.

The two lesions, both required for the answer:

- ① **CAECA** — thickened ulcerated wall containing a **hard, dry, laminated, yellow-white CAECAL CORE**
- ② **LIVER** — circular, depressed, yellow-grey to green necrotic foci up to 1–2 cm with a raised rim, the classic **"TARGET" or "BULL'S-EYE" lesion**

The name "blackhead" comes from cyanosis of the head, but this is **inconstant and NOT a reliable sign** — the liver lesion is.

19 · ECTOPARASITES OF POULTRY

Parasite	Effect
Lice — <i>Menopon gallinae</i> (shaft louse), <i>Menacanthus stramineus</i> (body louse)	Chewing lice , they feed on feather debris and scurf, not blood; cause intense irritation, restlessness, feather damage, reduced growth and egg production . Spend their entire life on the bird
Red mite — <i>Dermanyssus gallinae</i>	Blood-sucking and NOCTURNAL — it feeds at night and hides in cracks of the house by day , so the bird looks clean on daytime examination. Causes severe anaemia (pale comb), restlessness, reduced production and even death ; also spreads spirochaetosis. Treat the house , not only the bird
Northern fowl mite — <i>Ornithonyssus sylviarum</i>	Blood-sucker that lives on the bird permanently , around the vent; blackened, matted, dirty vent feathers
Scaly leg mite — <i>Knemidocoptes mutans</i>	Burrows under the leg scales → the scales lift, thicken and become crusty and deformed ("scaly leg") , lameness
Depluming mite — <i>Knemidocoptes gallinae</i>	Burrows at the feather base → intense itching and feather pulling and loss
Fowl tick — <i>Argas persicus</i>	Soft tick, nocturnal blood feeder hiding in cracks; anaemia, "tick worry", tick paralysis , and it is the vector of <i>Borrelia anserina</i>
Sticktight flea — <i>Echidnophaga gallinacea</i>	Attaches in clusters around the eyes, comb and wattles ; ulceration and anaemia

20 · AVIAN NUTRITIONAL DEFICIENCIES ★★★

Nutrient	The named lesion
Vitamin A	"NUTRITIONAL ROUP" — white pustular nodules in the pharynx, oesophagus and nasal passages (resembles wet fowl pox but has no Bollinger bodies); xerophthalmia with cheesy material under the eyelid; urate deposits in the kidney and ureters ; squamous metaplasia of glandular epithelium; poor growth and hatchability
Vitamin D₃ / Ca / P	RICKETS in chicks — soft rubbery beak and bones, "BEADED RIBS" (rachitic rosary) at the costochondral junctions, bent keel, inability to stand . In layers — thin-shelled and shell-less eggs, osteomalacia and drop in production
Vitamin E / Selenium ★	Three distinct syndromes, all worth marks: ① ENCEPHALOMALACIA ("CRAZY CHICK DISEASE") — ataxia, backward somersaults, head retraction; the CEREBELLUM is swollen, soft, haemorrhagic and yellow-green with necrosis ② EXUDATIVE DIATHESIS — capillary leakage giving greenish-blue gelatinous subcutaneous oedema over the BREAST and thighs ③ NUTRITIONAL MUSCULAR DYSTROPHY — white chalky streaks in the breast muscle and in the GIZZARD (gizzard myopathy in turkeys)
Vitamin K	Haemorrhagic syndrome — prolonged clotting, bleeding into muscle, subcutis and gut; worsened by sulpha drugs and mycotoxins
Riboflavin (B₂)	"CURLED-TOE PARALYSIS" — the chick walks on its hocks with the toes curled inward ; caused by peripheral nerve demyelination of the sciatic nerve , which becomes visibly swollen. Poor hatchability with clubbed down
Thiamine (B₁)	POLYNEURITIS — "STAR-GAZING" posture , the head retracted over the back (opisthotonus), leg paralysis
Niacin	Bowed legs and enlarged hock joints (resembles perosis but the Achilles tendon does not slip), inflamed mouth and tongue ("black tongue")
Manganese / Choline	PEROSIS ("SLIPPED TENDON") — the hock enlarges and flattens and the gastrocnemius tendon slips off the condyle , twisting the leg outward; also chondrodystrophy and short thick bones. Manganese deficiency also causes nutritional chondrodystrophy in the embryo
Biotin / Pantothenic acid	Dermatitis of the FOOT PADS and the CORNERS OF THE MOUTH and eyelids , cracked scaly feet, poor feathering; biotin deficiency also gives fatty liver and kidney syndrome
Zinc	Poor feathering with frizzled feathers, short thickened long bones, foot-pad dermatitis
Salt (excess)	Excess sodium → ascites, subcutaneous oedema, wet litter , kidney damage; deficiency → cannibalism

21 • MISCELLANEOUS CONDITIONS ★★★

Condition	What it is
Heat stroke	Indian summer, poor ventilation, high stocking. Panting, prostration, sudden death of the heaviest birds. PM: generalised congestion, pulmonary congestion and oedema, rapid onset of rigor and rapid decomposition ; the carcass spoils quickly
Vent gleet (cloacitis)	Chronic inflammation of the cloaca with a foul-smelling, pasty, yellow-white discharge that soils and bares the vent feathers; associated with poor hygiene, and it predisposes to vent pecking and cannibalism
INTERNAL LAYER	The ovum is ovulated but the infundibulum fails to pick it up , so the yolk drops free into the abdominal cavity → EGG-YOLK PERITONITIS : yellow, cheesy, foul yolk material plastered over the viscera, peritonitis, and death. Often follows Ranikhet disease, salpingitis or rough handling
FALSE LAYER ★	The bird looks and behaves like a layer with a normal comb and pelvis, but never lays an egg . It is caused by permanent OVIDUCT damage, classically by infectious bronchitis virus infecting the chick in its first two weeks . The ovary is active but the oviduct is blind, cystic and fluid-filled, so ovulated yolks fall into the abdomen
Pendulous crop	The crop becomes atonic, dilated and impacted , hanging as a heavy pendulous sac; sour smell; associated with fibrous feed, water deprivation followed by gorging, and <i>Candida</i>
Breast blister (keel bursitis)	Inflammation of the sternal bursa over the keel , from wet caked litter, poor perches and any condition that makes the bird sit — leg weakness, synovitis. Starts as a clear fluid-filled swelling, becomes a hard caseous mass. A major carcass-downgrading defect
ASCITES SYNDROME (pulmonary hypertension syndrome)	A metabolic disease of FAST-GROWING BROILERS : oxygen demand outstrips the capacity of the lung, made worse by cold, high altitude, poor ventilation and pellet feeding . Pulmonary hypertension → RIGHT VENTRICULAR HYPERTROPHY AND DILATATION → right heart failure . PM: abdomen distended with clear straw-coloured fluid plus YELLOW GELATINOUS FIBRIN CLOTS; a dilated flabby right ventricle; and a congested, firm, "nutmeg"/cardiac LIVER often coated with fibrin . Hydropericardium may also be present
FATTY LIVER AND KIDNEY SYNDROME (FLKS)	BIOTIN deficiency in fast-growing broilers of about 2–4 weeks on wheat-based low-biotin rations. Sudden death of apparently healthy chicks, often lying on the breast with the head thrown back. The LIVER and KIDNEYS are pale, swollen and fatty; hypoglycaemia is the fatal event; the heart is also pale

FATTY LIVER (haemorrhagic) SYNDROME

A quite different disease — of **OVERWEIGHT CAGED LAYERS** on a high-energy diet with no exercise. The **liver is grossly enlarged, yellow, greasy, soft and friable, and is easily torn**; large **subcapsular haemorrhages and blood clots** are present, and **fatal rupture with massive intra-abdominal haemorrhage** is the usual cause of death, together with a fall in egg production. Abundant abdominal fat

CAGE LAYER FATIGUE

Osteoporosis of high-producing caged layers — calcium and phosphorus are drained for shell formation while the bird gets **no exercise**, so structural bone is resorbed. The hen is **found lying on her side in the cage, unable to stand, but bright and eating; bones (ribs, vertebrae, femur) are soft and fracture easily**, and collapse of a thoracic vertebra causes paralysis. Shells become thin

GOUT ★

Deposition of **URATES** because birds are **uricotelic** and any renal failure or dehydration precipitates them. **VISCERAL gout** — a chalky white "frosting" of urate crystals on the **pericardium, liver capsule, air sacs and peritoneum**; kidneys and ureters distended with white urate. **ARTICULAR gout** — soft white "toothpaste-like" urate **tophi in and around the joints and tendon sheaths** of the feet and hocks, causing lameness. **Causes**: nephrotoxic damage (**ochratoxin**, nephropathogenic **IB virus**, avian nephritis virus), excess dietary **protein and calcium**, **water deprivation**, sodium bicarbonate excess, vitamin A deficiency. Micro: needle-shaped urate crystals with a surrounding **granulomatous "tophus"** reaction

Haemorrhagic syndrome

Aplastic anaemia and thrombocytopenia from mycotoxins (**trichothecenes, aflatoxin**), sulpha drugs or chicken infectious anaemia. **Pale, fatty, yellow bone marrow**; haemorrhages in muscle, subcutis, and viscera; pale carcass

Round heart disease

An idiopathic **cardiomyopathy** of turkeys and broilers; the heart is **globular, rounded, pale and flabby with a dilated thin-walled ventricle**; sudden death

Impaction of the oviduct · egg bound

A **retained egg or laminated caseous mass in the oviduct or cloaca**; the hen strains, and it may cause **cloacal or oviduct prolapse** and then **vent pecking**. Associated with the first large eggs of a young pullet, oversized eggs, hypocalcaemia and salpingitis

BUMBLE FOOT (pododermatitis)

Infection, usually ***Staphylococcus aureus***, entering a wound on the plantar footpad from **rough perches, wire floors or wet caked litter**. The foot is **hot, swollen and painful, with a dark scab, and contains a firm CASEOUS CORE** (remember — birds cannot make liquid pus). Leads to lameness, arthritis and osteomyelitis

22 · COMMON VICES

Cannibalism in its several forms — **toe picking** (chicks), **feather pecking** (growers), **vent picking** (layers, often after a prolapse — the most damaging), **head picking**, and **egg eating**.

Causes: overcrowding and high stocking density · **excessively bright light** · high temperature and poor ventilation · **nutritional deficiency of protein, METHIONINE, salt or fibre** · boredom and lack of feeder/drinker space · prolapse, vent gleet or ectoparasites drawing attention to the vent · mixing birds of unequal size.

Control: **beak trimming** · reduce light intensity or use **red light** · lower stocking density · balanced ration with adequate methionine and salt · provide roughage or pecking blocks · remove and treat injured birds at once.

23 · BLANK-FILL DRILL — UNIT 5

- The avian granulocyte is the **heterophil**, which lacks **myeloperoxidase**, so avian pus is **caseous**
- Chickens have **no** lymph nodes; B cells mature in the **bursa of Fabricius**
- Birds are **uricotelic**, which is why renal failure causes **gout**
- Ranikhet disease is caused by **avian paramyxovirus-1**
- R2B (Mukteswar) is a **mesogenic** strain; LaSota and F1 are **lentogenic**
- Haemorrhages on the tips of the **proventricular glands** and necrosis of the **caecal tonsils** = Ranikhet disease
- IBD virus is a **Birnavirus** and targets the **bursa of Fabricius**
- Paint-brush haemorrhages in thigh and breast muscle occur in **IBD**
- Chicken infectious anaemia gives **pale fatty (yellow)** bone marrow and **thymic** atrophy
- Blue wing disease is **gangrenous dermatitis** with chicken infectious anaemia
- Nephropathogenic strains of **infectious bronchitis** cause gout; early infection creates a **false layer**
- Blood-stained tracheal casts with syncytia and **intranuclear** inclusions = **ILT**
- Fowl pox inclusions are **Bollinger** bodies and are **intracytoplasmic**; spread is by **mosquitoes**
- The diphtheritic form of fowl pox kills by **suffocation**
- Marek's disease is caused by a **herpesvirus** and is a **T**-cell tumour; leukosis is a **retrovirus** and a **B**-cell tumour
- Enlarged asymmetrical **sciatic** nerves occur in **Marek's** disease, never in leukosis
- A nodular tumour in the bursa points to **lymphoid leukosis**
- Marek's disease spreads by **feather follicle dander** and is prevented by the **HVT** vaccine
- Lymphoid nodules in the **gizzard** muscle are diagnostic of **avian encephalomyelitis**
- Hydropericardium syndrome is also called **Angara disease** and is caused by fowl adenovirus serotype **4**
- Inclusion body hepatitis shows **basophilic intranuclear** inclusions in hepatocytes
- EDS-76 is caused by a **duck adenovirus** and produces **shell-less and depigmented** eggs
- Rupture of the **gastrocnemius** tendon follows **reovirus** arthritis
- The colibacillosis triad is **perihepatitis, pericarditis and airsacculitis**
- Mushy chick disease is **omphalitis / yolk sac infection**

- Foul-smelling nasal discharge with facial swelling = **infectious coryza**
- Turkish-towel pseudomembrane in the small intestine = **necrotic enteritis**, predisposed by **coccidiosis**
- White diarrhoea with a pasted vent in chicks = **pullorum disease**
- A **bronze**-coloured enlarged liver in an adult bird = **fowl typhoid**
- Pullorum and fowl typhoid spread mainly **vertically** and are controlled by the **whole-blood plate agglutination** test
- Sudden death with cyanotic wattles and pinpoint hepatic necrosis = **fowl cholera**; organism is **bipolar**
- Avian tuberculosis shows **little or no calcification**, unlike mammalian TB
- Avian spirochaetosis is caused by **Borrelia anserina**, transmitted by **Argas persicus**
- Chronic respiratory disease is caused by **Mycoplasma gallisepticum**; infectious synovitis by **M. synoviae**
- Chlamydia psittaci causes **psittacosis / ornithosis** and is a **zoonosis**
- Brooder pneumonia is caused by **Aspergillus fumigatus**, whose hyphae are **septate**
- White comb (favus) is caused by **Microsporium gallinae**
- Ducklings are the most susceptible species to **aflatoxin**
- Ochratoxin damages the **kidney** and causes **gout**; T-2 toxin causes **oral lesions** at the beak commissure
- Heterakis gallinarum transmits **Histomonas meleagridis**
- Syngamus trachea is the **gapeworm** and lives in the **trachea**
- Eimeria tenella affects the **caeca** and forms **caecal cores**
- Ladder-like white lesions in the duodenum = **Eimeria acervulina**
- Salt-and-pepper petechiae in the mid intestine = **Eimeria necatrix**
- Target or bull's-eye liver lesions with caecal cores = **histomoniasis (blackhead)**, worst in **turkeys**
- Dermanyssus gallinae is the **red mite**, feeds at **night** and hides in **cracks**
- Knemidocoptes mutans causes **scaly leg**
- Nutritional roup is **vitamin A** deficiency
- Crazy chick disease (encephalomalacia) affects the **cerebellum** and is **vitamin E** deficiency
- Exudative diathesis gives **greenish-blue** subcutaneous oedema over the **breast**
- Curled-toe paralysis = **riboflavin** deficiency; star-gazing = **thiamine** deficiency
- Perosis (slipped tendon) = **manganese and choline** deficiency
- A bird that looks like a layer but never lays is a **false layer**, usually caused by **infectious bronchitis**
- Ascites syndrome of broilers is caused by **pulmonary hypertension** and gives **right ventricular** hypertrophy
- Fatty liver and kidney syndrome is due to **biotin** deficiency and kills by **hypoglycaemia**
- Fatty liver haemorrhagic syndrome occurs in **overweight caged layers** and kills by liver **rupture**
- Cage layer fatigue is **osteoporosis** from calcium drain plus lack of **exercise**
- Urate frosting on the pericardium and liver capsule = **visceral gout**
- Bumble foot is usually caused by **Staphylococcus aureus** and contains a **caseous core**
- Cannibalism is controlled by **beak trimming** and by reducing **light intensity**